Luke Eyles

Mechanical and Mechatronics Engineering Student
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Experience

Jenkins Engineering Defence Systems Aug 2019 – Feb 2020
Engineering Intern in Research and Development Department

- Created Signal Generator GUI application in Matlab that interfaced with software defined radios to transmit pre-set or user defined signals.
- Developed Signal Analyser GUI in Matlab, allowing a quick overview of signal characteristics at a glance, or in depth analysis.
- Created system to record sensor data from an IMU to a single board computer using C++.

University of Technology Sydney Mar 2018 – Jun 2018
Assistant Lab Tutor for 48510 Introduction to Electrical Engineering

- Answered students' questions about the signal generators and oscilloscopes used.
- Explained electrical concepts and components.
- Debugged students' circuits and walked through how to fix the error.

Fusion Op Shop Hornsby Jan 2016 – Jan 2017
Weekly Volunteer

Education

University of Technology Sydney 2017 – 2022 (Expected)
Bachelor of Mechanical and Mechatronic Engineering,
Diploma in Professional Engineering Practice

- GPA: 6.43/7.00, High Distinction average.
- Robotics: Programmed two simulated pick and place robots to work together to build a wall in Matlab.
- Programming for Mechatronic Systems: Programmed a simulated aircraft in C++ to intercept enemy aircraft by predicting their trajectory.
- Mechatronics 2: Built an Arduino security system using a distance sensor mounted on a motor, capable of mapping a room and detecting intruders.
- Mechanical Design 1: Designed a robot to attempt the 2019 Warman Design and Build Challenge as part of a team.

- Manufacturing Engineering: Designed and manufactured a hand pump as part of a team.

Bradfield Senior College 2014 – 2016
Higher School Certificate

Skills

Technical Skills

- **C++:** Object oriented programming, using threads to process and send multiple streams of data, compiling using CMake, version control using Git.
- **Matlab**: Image processing, signal processing, robotics using Peter Corke's Robotics Toolbox and ROS Toolbox, GUI programming using App Designer.
- **Robot Operating System (ROS)**: Controlling real and simulated robots, processing incoming messages from sensors using both Matlab and C++.
- **Microcontrollers**: Interfacing with sensors, controlling LCD screen and motors, using timers.
- **Git**: Version control and collaboration.
- **CAD***:*
    - **Solidworks**: Designing parts, performing FEA, creating assemblies and technical drawings.
    - **Blender**: Creating meshes, using modifiers, applying procedural and image based textures.
- **Engineering workshops:** Using power tools, following safety procedures.
- **Soldering**
- **Python**: Machine learning using Keras with TensorFlow.
- **Microsoft Excel**
- **Microsoft Office**

Core Skills

- **Teamwork and team leadership:**
    - Dividing work and delegating between team members, setting deadlines for review and final submission, contributing to team meetings, ensuring all work is cohesive and completed to a high standard.
    - Evidenced by high total marks and high ratings by peers in university group projects.
- **Visual communication skills**:
    - Drawing up diagrams to communicate concepts to others clearly.
- **Written and verbal communication skills:**
    - Writing clear and concise reports, tailored to the audience.
    - Listening carefully to specifications and asking questions when clarification or help is needed.
    - Participating in team meetings by contributing new ideas and adding to others' ideas.